PROGRAM

circular linked list

```
#include <stdio.h>
#include <malloc.h>
struct node {
  int data;
  struct node* next;
};
struct node *head,*tail,*newnode,*temp,*prev;
void createnode(int d)
{
        newnode = (struct node*)malloc(sizeof(struct node));
        newnode->data = d;
        newnode->next = NULL;
}
void insert(int key,int data)
{
        int found = 0;
        temp = head;
    do
    {
     if(temp->data == key)
     {
        found = 1;
        break;
                }
        temp = temp->next;
      }while(temp != head);

      if(found == 0)
      {
              printf("key element not found \n");
      }
      else
      {
              createnode(data);
              if(temp == tail)
              {
                      temp->next = newnode;
                      tail = newnode;
                      tail->next = head;
              }
              else
              {
                      newnode->next = temp->next;
                      temp->next = newnode;
```

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                }
            }
        printf("inserted :%d after :%d",data,key);
}
void prepend(int d)
{
        createnode(d);
        if(head == NULL)
        {
                head = tail = newnode;
                tail->next = head;
        }
        else
        {
          newnode->next = head;
          head = newnode;
          tail->next = head;
        }
        printf("data prepended :%d \n",d);
}
void append(int d)
{
        createnode(d);

        if(head == NULL)
        {
                head = tail = newnode;
                tail->next = head;
        }
        else
        {
                tail->next = newnode;
                tail = newnode;
                tail->next = head;
        }
        printf("data appended :%d \n",d);
}
void display()
{
    temp = head;
    do
    {
        printf("%d ->",temp->data);
        temp = temp->next;
          }while(temp != head);
}
```

```
void delete(int data)
{
        int found = 0;
        temp = head;
    do
    {
     if(temp->data == data)
     {
        found = 1;
        break;
                }
        temp = temp->next;
         }while(temp != head);

         if(found == 0)
         {
                printf("key element not found \n");
         }
         else
         {
                if(temp == head)
                {
                        head = head->next;
                        tail->next = head;
                        free(temp);
                }
                printf("data deleted :%d \n",data);
         }
}
int main()
{
        append(10);
        append(20);
        append(30);
        prepend(40);
        display();

        printf("\n--\n");
        insert(20,25);
        display();
        printf("\n--\n");
        delete(10);
        display();
        printf("\n--\n");
        display();
}
```

OUTPUT:

data appended :10

data appended :20

data appended :30

data prepended :40

40 ->10 ->20 ->30 ->
--
inserted: 25 after:20
40 ->10 ->20 ->25 ->30 ->
--
data deleted :10

40 ->20 ->25 ->30 ->
--